Hai Duong

Fairfax, VA | (+1) 571-340-5970 | hduong22@gmu.edu | Personal Website | Google Scholar

SUMMARY

Applied Scientist Intern at AWS and PhD student in Computer Science at George Mason University specializing in automated reasoning, and scalable verification of deep neural networks. Lead developer of NeuralSAT, an open-source engine ranked 2nd at VNN-COMPs, with published papers at CVPR, NeurIPS, CAV, and FSE.

EDUCATION

- **PhD**, Computer Science, George Mason University, Fairfax, VA 2022–present
- **MS**, Electrical Engineering, Hanoi University of Science and Technology, Hanoi, VN 2021

EXPERIENCE

- **Applied Scientist**, Prime Air, Amazon, *Seattle, WA* 2026 (expected)
- **Applied Scientist Intern**, Amazon Web Services, Amazon, *Santa Clara, CA* 2026
  - Developed SMT-based path enumeration and fuzzing for AWS compliance, achieving 100% test coverage and exposing legacy-migration divergences.
  - Built automated Java/Python to ControlsDSL migration framework with bidirectional equivalence verification, porting 40% of staging rules to production.
- **Graduate Researcher**, ROARS Lab, George Mason University, *Fairfax, VA* 2022–present
  - Architected NeuralSAT (Python, PyTorch, CUDA), an open-source GPU-accelerated DNN verification engine; ranked 4th (2023, 2026) and 2nd (2024, 2025) at VNN-COMP.
  - Developed VeriStable (FSE'24), a DPLL-based verifier exploiting stable neurons; outperforms α-β-CROWN and NeuralSAT and solved 48 of 51 hard GDVB problems vs. 4 for the baseline (12× more).
  - Extended robustness verification to new perturbation classes with VeriS (FSE'26) and VeriDou (CVPR'26 Highlight): VeriS verifies 78% of 5,508 structural-robustness problems prior tools could not, while VeriDou exposes adversarial examples over 14,340 instances on networks prior methods claimed robust.
  - Designed APTP (NeurIPS'25), a verifier-independent proof format whose checker outperforms prior work across 16 networks and 400 problems.
  - Developed CLAD, a constrained abstract domain using a Lagrangian primal–dual method; over 8,640 instances, verifies up to 19% more properties than the best prior abstraction and 31% more on constrained properties at 5.7× the speed.
  - Built VerCa, an autoencoder robustness verifier solving 87% of latent-space properties, with zero invalid counterexamples.
  - Built ZonoGPT, an abstract domain for verifying large GPT models, up to GPT-2 medium with 300M+ parameters.
  - Built mBFV, a bound propagation method handling multi-parameter bit-flip perturbations.

SKILLS

- **Programming Languages:** Python, C/C++, MATLAB, SMTLib, Lean, PyTorch, CUDA, NumPy
- **Methods:** SAT/SMT solving, abstract interpretation, branch-and-bound, LP/MILP, reinforcement learning

SELECTED PUBLICATIONS (FULL LIST)

P1 Duong et al., *"Verifying Neural Networks with Reinforcement Learning"*, NeurIPS 2026.
P2 Song et al., *"LeanVIPR: A Formally Verified Certificate Checker for Exact MIP (Workshop)"*, NeurIPS 2026.
P3 Duong et al., *"Verifying Neural Network Robustness with Dual Perturbations"*, CVPR 2026.
P4 Duong et al., *"Verifying structural robustness of deep neural network"*, FSE 2026.
P5 Duong et al., *"Compositional neural network verification via assume-guarantee reasoning"*, NeurIPS 2025.
P6 Duong et al., *"Generating and checking dnn verification proofs"*, NeurIPS 2025.
P7 Duong et al., *"Neuralsat: A high-performance verification tool for deep neural networks"*, CAV 2025.
P8 Duong et al., *"Harnessing neuron stability to improve dnn verification"*, FSE 2024.
P9 Xu et al., *"Training for verification: Increasing neuron stability to scale dnn verification"*, TACAS 2024.

AWARDS

- High Impact Grant, George Mason University 2027
- Tinker Research Grants ($5,000 API Credits), Thinking Machines 2026
- Outstanding PhD Student Award, George Mason University 2026
- Highlight Paper Award [P3], CVPR 2026
- Spotlight Paper Award [P5], NeurIPS 2025